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Course Number	MEC 322
Course Title	Manufacturing Fundamentals
Semester/Year	Winter/2025
Section Number	08
Group Number	03

Assignment No.1

Assignment Title	Final Project Group Report
Submission Date	Apr 4, 2025
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1. Abstract

Over the course of this past semester, our group has designed and built a contraption that can pick up and place small balls with a relatively high degree of accuracy. Over the course of the design process, the device was reworked constantly until a specific design was settled on. This design used many different parts from everyone's individual designs in order to achieve the highest degree of success possible. Upon receiving the necessary materials, the device was promptly assembled and tested. These test runs proved successful as the results seemed positive. However, the device was not perfect. It had some minor flaws in its design which made its usage inconsistent and even difficult. As such, some workarounds were necessary, such as lowering the number of bars used as well as increasing the number of elastic bands that were used to close the claw of the device. While the device is not perfect, it still proved to be relatively successful when it came to its primary job of picking up and moving small spheres.

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3. Introduction

The purpose of this project is to create a four bar linkage [1] machine that can effectively pick up balls and transport them to a box a distance away and deposit them. The machine was to be created from a 8x10 inch sheet of acrylic, which was laser cut to create our parts. We were given other materials to work with, such as screws, nuts, rubber bands, paper clips and twine. Our machine uses all of these parts in its design. The design process went through 2 prototype stages, in which the machines were simulated through CAD, popsicle sticks and cardboard (view figure 6.1) to test their geometry.

After the ideas were brainstormed, the cut files for our semi-final designs were submitted. Once they were finished one of our team members brought it home to assemble it. The machine joints were joined with screws and nuts, while the claw was glued together to remain static and level. It was noticed that during use the nuts would become unscrewed due to friction from the bars, so they were thread locked in optimal positions using a little glue.

To close and open the claw, spring force equations were used to figure out the optimal angle for the tension in the rope. Using those equations and the size of the balls, the rubber band and string were placed on the claw accordingly.

The semi-final design was tested in the lab, and was deemed not good enough. I was too long and too unpredictable to guarantee good results. This led to the synthesis of the final design, which can be seen in our CAD drawings.

4. CAD Drawing

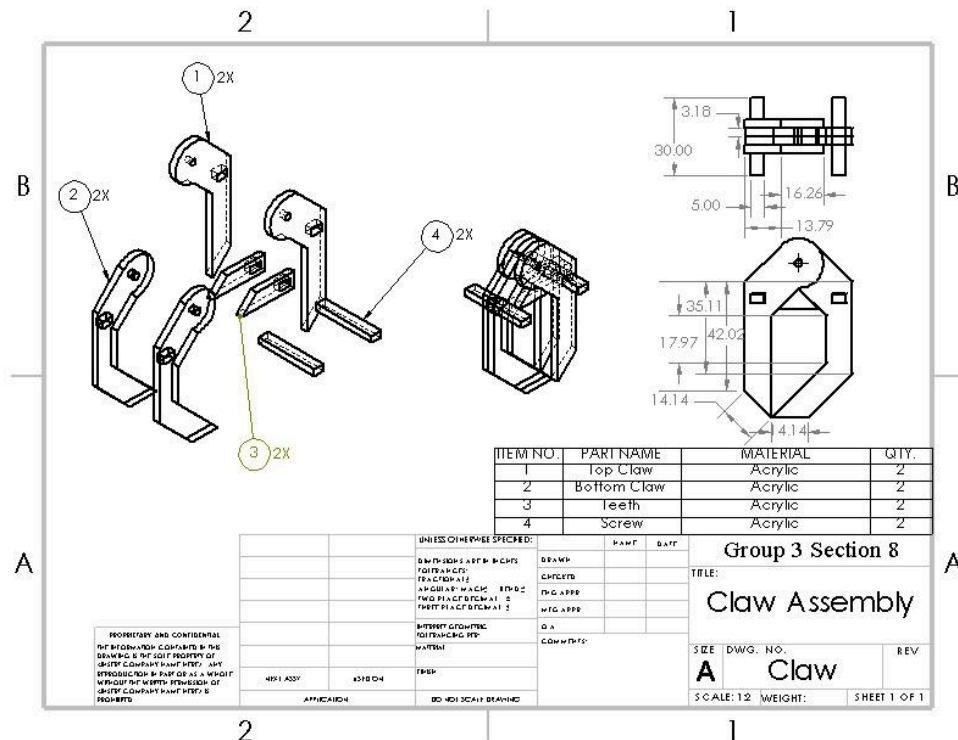


Figure 4.1 - Claw Assembly CAD Drawing

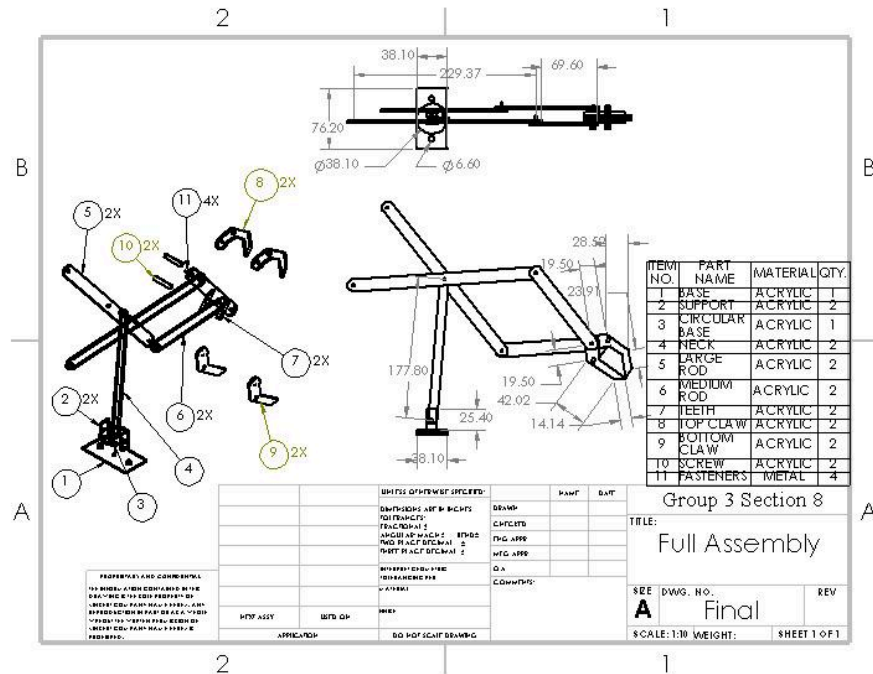


Figure 4.2 - Full Assembly CAD Drawing

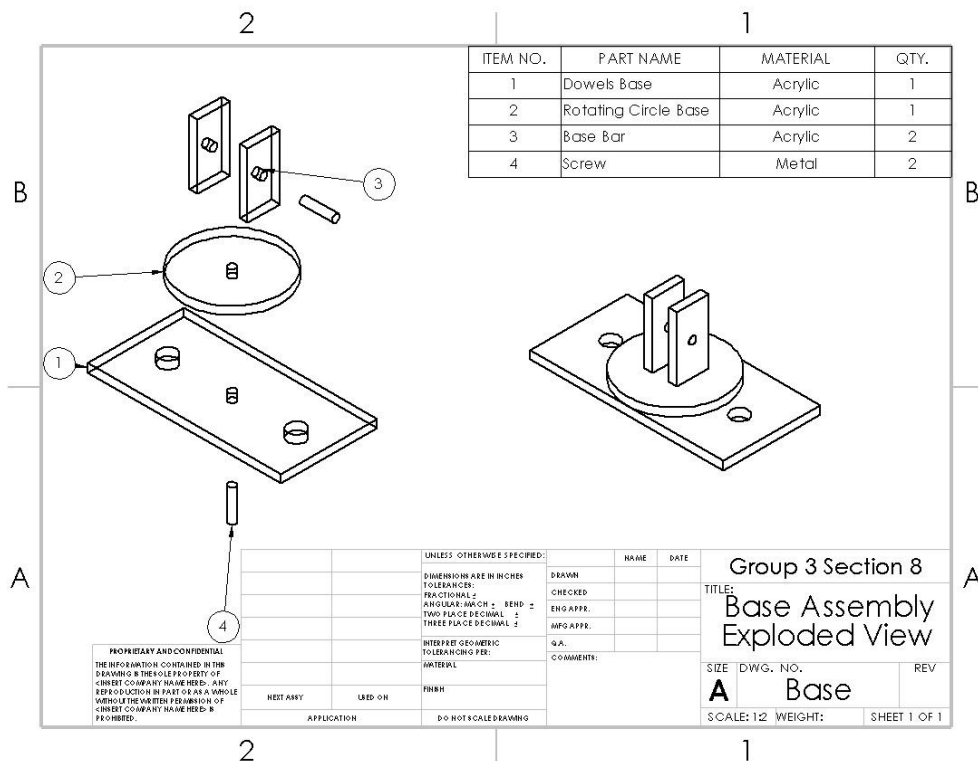


Figure 4.3 - Base Assembly Exploded View CAD Drawing

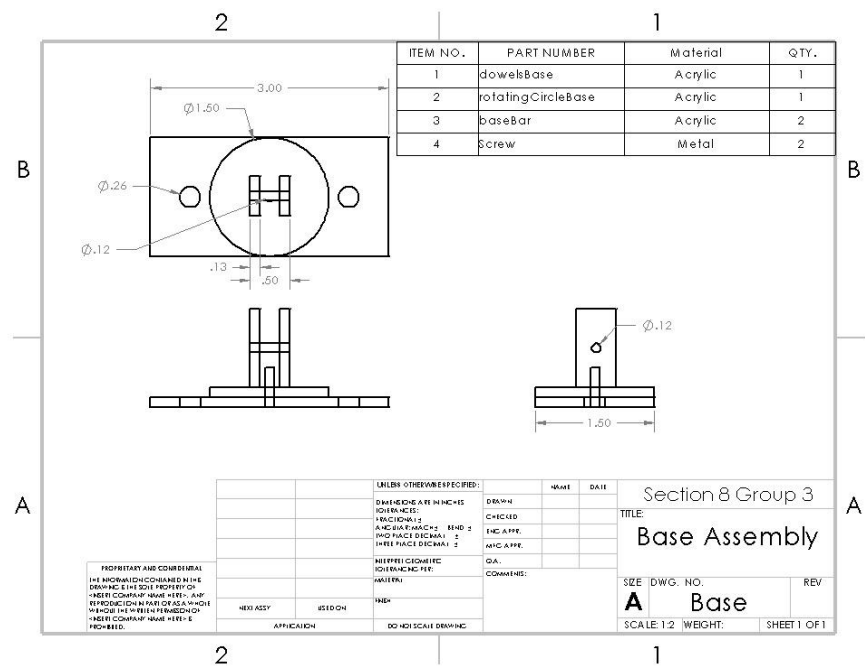


Figure 4.4 Base Assembly CAD Drawing

5. References

[1] M. West, “Four-Bar Linkages,” *dynref.engr.illinois.edu*, 2015.
<https://dynref.engr.illinois.edu/aml.html>

6. Appendix

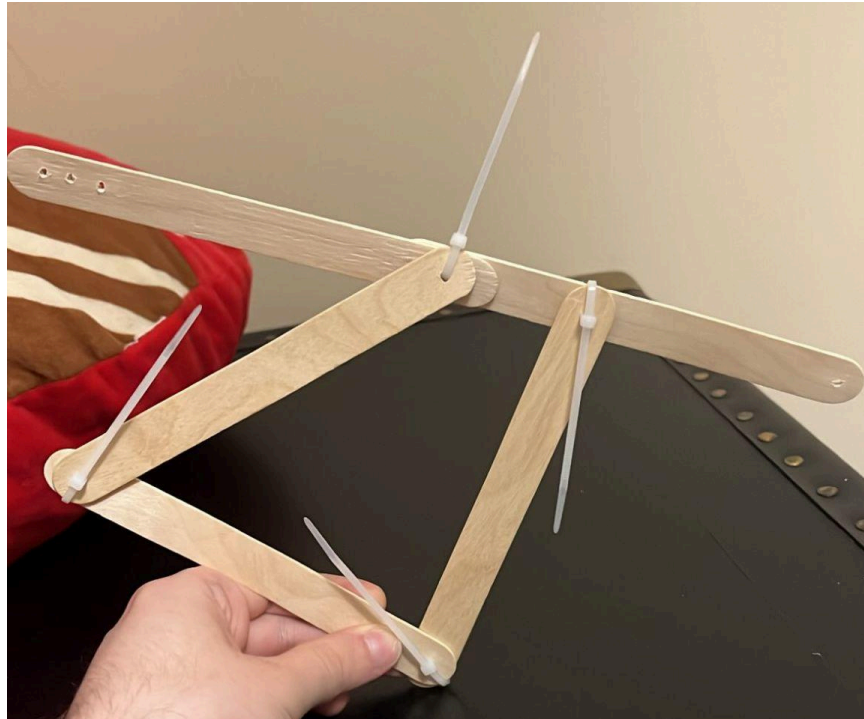


Figure 6.1 Image of Prototype made with popsicle sticks